ARTIN'S THEOREM ON CONTRACTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6502-6530 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 6532-6545, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 11 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 41 of queue_artin.json (pb=672, stmt 6502-6530, proof 6532-6545).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Editorial note. Statement carries a reference/slogan/history environment (patchenv -> patch-sd-envs.tex).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original theorem 0GHF: sources/stacks-artin/pinned-original-source.tex, source lines 5613–5617. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0GI7: sources/stacks-artin/pinned-original-source.tex, source lines 6212–6215. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0GI8: sources/stacks-artin/pinned-original-source.tex, source lines 6237–6240. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0GHC: sources/stacks-artin/pinned-original-source.tex, source lines 5372–5377. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original theorem 0GI5: sources/stacks-artin/pinned-original-source.tex, source lines 5636–5640. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0GIA: sources/stacks-artin/pinned-original-source.tex, source lines 6452–6455. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0GI6: sources/stacks-artin/pinned-original-source.tex, source lines 6175–6178. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 04FH: sources/stacks-artin/pinned-original-source.tex, source lines 5625–5629. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0GE7: sources/stacks-artin/pinned-original-source.tex, source lines 5072–5083. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0GIC: sources/stacks-artin/pinned-original-source.tex, source lines 6548–6569. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0GDK: sources/stacks-artin/pinned-original-source.tex, source lines 6545–6545. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1. *Let S be a locally Noetherian scheme such that $\mathcal{O}_{S,s}$ is a G-ring for all finite type points $s \in S$. Let X' be an algebraic space locally of finite type over S. Let $T' \subset |X'|$ be a closed subset. Let W be a locally Noetherian formal algebraic space over S with W_{red} locally of finite type over S. Finally, we let*

$$g : X'_{/T'} \longrightarrow W$$

be a formal modification, see Algebraization of Formal Spaces, Definition 0GDK. If X' and W are separated[1] *over S, then there exists a proper morphism $f : X' \to X$ of algebraic spaces over S, a closed subset $T \subset |X|$, and an isomorphism $a : X_{/T} \to W$ of formal algebraic spaces such that*

1. *T' is the inverse image of T by $|f| : |X'| \to |X|$,*
2. *$f : X' \to X$ maps $X' \setminus T'$ isomorphically to $X \setminus T$, and*
3. *$g = a \circ f_{/T}$ where $f_{/T} : X'_{/T'} \to X_{/T}$ is the induced morphism.*

In other words, $(f : X' \to X, T, a)$ is a solution as defined earlier in this section.

Proof. Let F be the functor constructed using X', T', W, g in this section. By Lemma 0GHC it suffices to show that F corresponds to an algebraic space X locally of finite type over S. In order to do this, we will apply Proposition 0GE7. Namely, by Lemma 0GHF the diagonal of F is representable by closed immersions and by Lemmas 04FH, 0GI5, 0GI6, 0GI7, 0GI8, and 0GIA we have axioms [0], [1], [2], [3], [4], and [5]. □

[1] See Remark 0GIC.